NOT MEASUREMENT SENSITIVE

MIL-B-43049D
20 November 1992
SUPERSEDING
MIL-B-43049C(IH)
30 September 1975

MILITARY SPECIFICATION

BACKING PLATES, INSIGNIA, BRANCH OF SERVICE AND CAP, INFANTRY, ARMY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the plastic backing plates used by Army Infantry enlisted personnel (see 6.1).

1.2 Classification. Backing plates shall be of the following types (see 6.2).

Type I - Backing Plate, Insignia, Service Cap, Infantry Blue
Type II - Backing Plate, Insignia, Branch of Service, Collar, Infantry Blue

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

NN-P-71 - Pallets, Material Handling Wood Stringer Construction 2-Way and 4-Way (Partial)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in preparing this document should be addressed to: Director, The Institute of Heraldry, US Army, Cameron Station, Alexandria, VA 22304-5050, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

AMSC N/A FSC 8455
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MILITARY

MIL-P-15011 - Pallets, Material Handling, Wood Post Construction, 4-Way Entry

STANDARDS

FEDERAL

FED-STD-595 - Colors Used In Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Document Order Desk, Building 4D, Section D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

THE INSTITUTE OF HERALDRY (TIOH)

B-2-51 - Backing Plate, Insignia, Cap & Collar, U.S. Army Infantry

(Copies of specifications, standards, drawings and specification sheets required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

(Figure 1 is a miniature copy of an Institute of Heraldry drawing and is for information only.)

2.2 Non-Government publications. The following documents form a part of this document to the extend specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-570 - Plastic, Water Absorption of
ASTM-D-756 - Determination of Weight & Shape Changes of Plastics Under Accelerated Service Conditions
ASTM-D-3951 - Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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THE COLOR ASSOCIATION OF THE UNITED STATES

The Standard Color Reference of America, Tenth Edition

United States Army Color Card, The Official Colors for Branches, Agencies, etc.

(Application for copies of the color cards should be addressed to the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-1187.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications or MS standards), the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Standard sample. Unless otherwise specified (see 6.2), the contracting officer shall furnish a standard sample of the backing plate which shall be used for matching color and finish only. Variation from this specification may appear in the standard sample; however, in such cases the specification shall govern.

3.3 Materials. Materials shall be as specified herein.

3.3.1 Plastic. The backing plate shall be molded from virgin plastic material. Cellulose Nitrate or Cellulose Acetate Butyrate shall not be used.

3.4 Construction. The backing plate, Type I or Type II, shall be molded in accordance with the details and dimensions of TIOH Drawing B-2-51 (see Figure 1). The backing plates shall meet the performance requirements specified herein.

3.4.1 Stability. When tested as specified in 4.7.1, the backing plate shall not twist, chip, crack, break or blister and shall be free from warp. When the backing plate is placed on a surface plate with the beveled edge upward, at least 75 percent of the perimeter of the backing plate shall be in contact with the plate and there shall be no gap exceeding 1/32 inch (0.08 cm).

3.4.2 Water absorption. When tested as specified in 4.7.2, the water absorption shall not be greater than 2.5 percent.

3.4.3 Color. The color of Type I and Type II backing plates shall be light blue approximating Chip No. 15177 from FED-STD-595 or the standard sample (See 3.2).

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3.4.4 Knockout pin mark. Knockout pin marks and breakoff marks shall be trimmed flush within 1/64 inch (0.04 cm) of adjacent areas. There shall be no sharp edges. The marks shall not be located in areas which are visible under actual use.

3.5 Finish. The backing plates shall be free of warp, twist, cracks, blisters, scratches, sink marks, flash, fins and roughness at breakoff point and shall meet the quality provisions as specified herein. The occurrence of defects shall not exceed the applicable acceptable quality levels.

3.6 Workmanship. The backing plates shall be free of warp, twist, cracks, blisters, scratches, sink marks, flash, fins and roughness at breakoff point and shall meet the acceptable quality provisions established by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, not does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5 and 4.6.

4.4 First article sample inspection. Inspection and testing of the first article sample (see 3.2), shall be made of a completely finished backing plate for all provisions of this specification applicable to the end product examination and tests.

4.5 Quality conformance inspection. Inspection shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated (see 6.2).

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4.5.1 Inspection of components.

4.5.1.1 Certification of compliance. The plastic material may be accepted on the basis of the contractor's certificate of compliance that the plastic material meets the requirements of 3.3.1.

4.5.2 Inspection of the end item.

4.5.2.1 Visual examination of the backing plate. Examination shall be made at a distance of approximately 16 to 22 inches. The defects found during examination shall be classified in accordance with 4.5.2.1.1 and 4.5.2.1.2. The lots shall be inspected in accordance with 4.5.2.1.3. The unit of product for the examinations in 4.5.2.1.1 and 4.5.2.1.2 shall be one completely finished backing plate.

4.5.2.1.1 Examination of backing plate for defects in color, finish, design, construction and workmanship. Defects designated by an asterisk (*) shall be classified as major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously.

TABLE I. DEFECTS

EXAMINE	DEFECT	CLASSIFICATION Major	(*)	Minor
Color and finish	Does not match standard sample or approximate FED-STD-595 Chip No. 15177	x	-	-
	Not smooth	x	-	-
	Not uniform shade	x	-	-
	Beveled section does not have a high uniform luster	x	-	-
	Discoloration such as burn, spot, stain, streak or speck clearly noticeable	x	-	-
	Any imbedded foreign matter	-	*	-
	Any crack, break, blister, lump, pit, crater, or other imperfection clearly noticeable	x	-	-
	Any scratch on beveled surface	-	*	-
Workmanship	Any chip, nick, dent, dig, gouge, sink mark, short, flash, fin, roughness or burr clearly noticeable	-	-	x
	Any warp, twist, unevenness or distortion	-	-	x
	Any hole not formed as indicated on drawing	-	*	-
	Knockout pin or breakoff mark located in areas visible when in actual use	x	-	-
	Not trimmed within 1/64 inch (0.04 cm) of adjacent areas	-	*	-
	Edge of hole not clean or not smooth, i.e., any burr or flash fin	-	*	-
	Breakoff mark not neatly accomplished, i.e., sharp or rough projection which is clearly noticeable or could possibly abrade cloth material on which worn	x	-	-

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4.5.2.1.2 Examination of the backing plate for defects in dimensions. Any dimension which is not within the specified tolerance shall be classified as a defect. The sample shall be one completely finished backing plate from each mold cavity.

4.5.2.1.3 Inspection levels and acceptable quality levels (AQL's). The inspection levels and the acceptable quality levels (AQL's) expressed in defects per hundred units shall be as follows:

	Inspection Level	AQL Major	AQL Total
For defects applicable to 4.5.2.1.1	II	(one class)	4.0
For defects applicable to 4.5.2.1.2	S-1	(one class)	4.0

4.5.3 Examination for count of backing plates for intermediate containers. Backing plates packaged for shipment shall be examined to determine conformance with package markings and specified quantity. The sample unit for this examination shall be one box (intermediate container). The lot size shall be the number of intermediate containers. Any box containing less than the specified marked quantity of backing plates shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects per sample unit.

4.5.4 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

EXAMINE	DEFECT
Marking (exterior and interior)	Omitted, incorrect, illegible, or improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping or taping, inadequate strapping. Bulging or distortion of containers.
Content	Number of intermediate packages is more or less than required.
Weight	Net weight exceeds requirements

4.6 End item testing. Testing of the completely fabricated backing plates shall be performed in accordance with Table I for the characteristics shown therein.

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The sample unit shall be three completely fabricated backing plates of each type. One determination per sample unit shall be performed for each characteristic. The requirements are applicable to the individual unit. The inspection level shall be S-1 and AQL shall be 4.0 defects per hundred units.

4.7 Tests.

4.7.1 Stability at high temperatures and humidity. The backing plate shall be tested in accordance with ASTM-D-756 and shall then be examined to determine compliance with 3.4.1.

4.7.2 Test for water absorption. The backing plate shall be tested in accordance with ASTM-D-570 and shall then be examined to determine compliance with 3.4.2. Dimensional changes in the backing plate shall not be considered and reported.

4.8 Packaging Inspection. The preservation, marking and marking shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be Level C or Commercial, as specified (see 6.2).

5.1.1 Level C. Backing plates of one type only shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity. the supplier may use his own standard practice when it meets these requirements.

5.1.2 Commercial. Backing plates packaged as specified in 5.1 shall be packed in accordance with the applicable requirements of ASTM-D-3951.

5.2 Packing. Packing shall be Level C or Commercial as specified (see 6.2).

5.2.1 Level C. Backing plates of one type only shall be packed in a manner to insure carrier acceptance and safe delivery to destination at the lowest transportation rates for such supplies. Containers shall be in accordance with the rules and regulations of carriers applicable to the mode of transportation.

5.2.2 Commercial. Backing plates packaged as specified in 5.1 shall be packed in accordance with the applicable requirements of ASTM-D-3951.

5.3 Palletization. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be palletized on a 4-way entry pallet in accordance with load type 1A of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, Type I, Class 1, Style 1, Size A, Wood Group I, II, III or IV of MIL-P-15011, or, 4-way, Type IV, V or VIII, Class 1, Style A, Size 2, Wood Group I, II, III or IV, Grade A of NN-P-71. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

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5.4 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.5 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.8.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Type I and Type II backing plates covered by this specification are intended for the use of enlisted infantry personnel to mount their cap and branch of service insignia.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. When a first article is not required (see 3.1).

d. Whether a standard sample is to be furnished (see 3.2).

e. Drawing (see 2.1.2 and 3.4).

f. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

g. When palletization is required (see 5.3).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first production sample or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The first article should consist of one completed backing plate. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Standard sample. For access to standard samples, address the contracting activity issuing the invitation to bid (see 3.2).

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6.5 Subject term (key word) listing.

Plastic disk
Disk
Plate

6.6. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - IH

Review activities:
DLA - CT

Preparing activity:
Army - IH

(Project No. 8455-0694)

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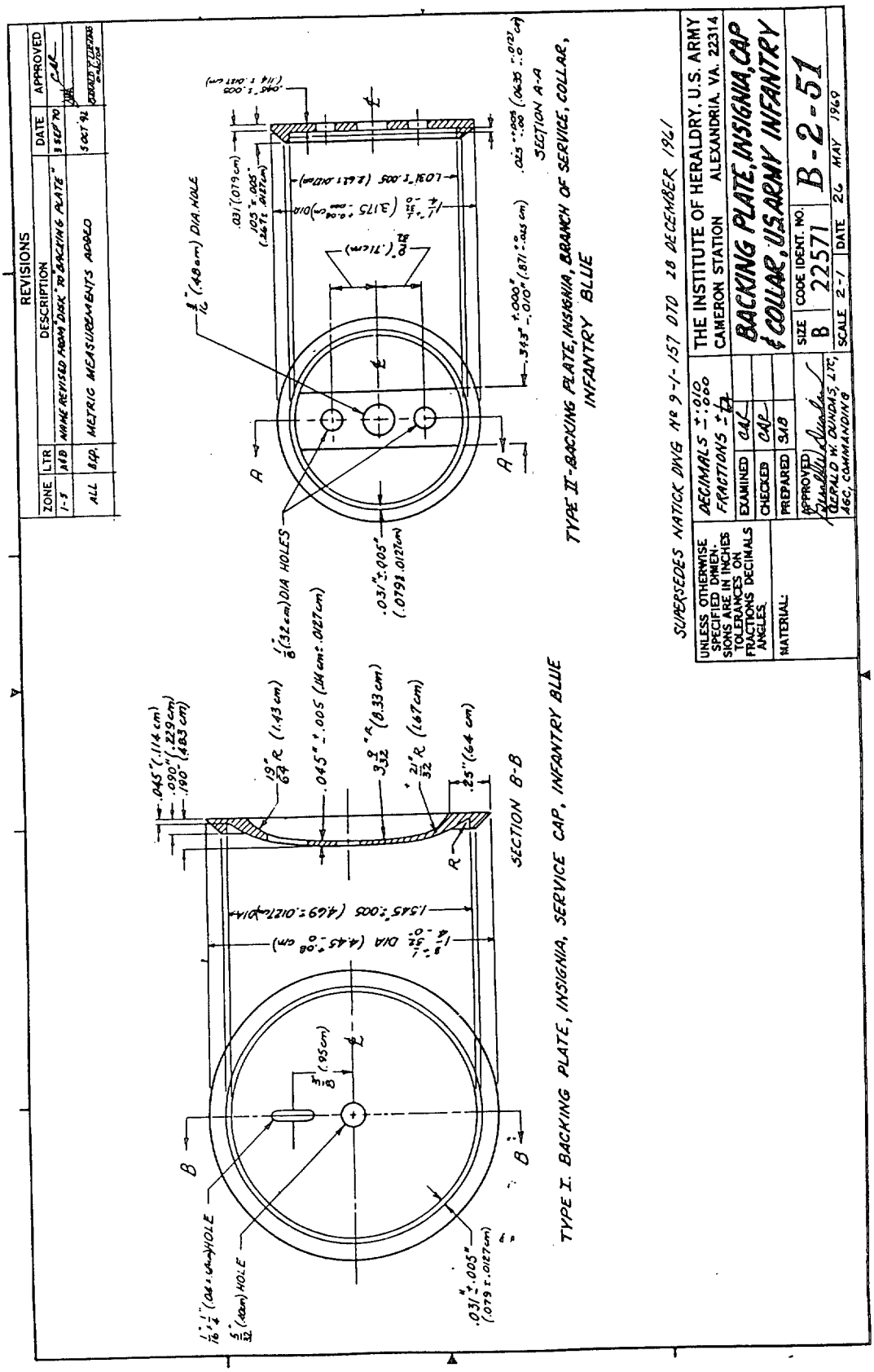


FIGURE 1. Backing Plate, Insignia, Cap & Collar, U.S. Army Infantry

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-43049D	2. DOCUMENT DATE (YYMMDD) 20 November 1992

3. DOCUMENT TITLE
BACKING PLATES, INSIGNIA, BRANCH OF SERVICE AND CAP, INFANTRY, ARMY

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
The Institute of Heraldry	(202) 274-6636	284-6636
c. ADDRESS (Include Zip Code) Bldg 15, Technical & Production Division Cameron Station Alexandria, VA 22304-5050	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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